National University of Computer and Emerging Sciences
Karachi Campus

Artificial Intelligence (AI2002)

Date: April 4th 2024

Course Instructor(s)

Dr. Fahad Sherwani, Dr. Farrukh Hassan Syed,
Dr. Maria Siddiqua, Mr. Syed Faisal Ali,
Mr. Syed Farooq Ahmed Zaidi,
Ms. Saeeda Kanwal, Ms. Fizza Aqeel,
Ms. Mafaza Mohi, Mr. Sandesh Kumar

Sessional-II Exam

Total Time (Hrs):	**1**
Total Marks:	**15**
Total Questions:	**3**

______________ ______________ ____________________________

Roll No Section Student Signature

Attempt all the questions.

CLO # 3: To demonstrate understanding and ability to implement the major concepts, approaches and research in evolutionary algorithms and CSP.

Q1: **[Time: 15 Minutes] [4+1= 5 Marks]**

A. Maximize the function $f(x) = x^2 + 2x - 1$, where x value ranges from [0-31]. Use genetic algorithm for finding the solution. Your solution must obey the following conditions:
   1. Initially you can select four chromosomes with the values: 8, 13, 19, and 23.
   2. Perform single point cross over at point 2 and 4.
   3. Perform mutation only on the weakest chromosome at point 3.
   4. State whether your answer is converging or diverging.

B. How does mutation enhance the genetic variation within a population?

CLO # 2: To identify and relate methods of search and practically apply the corresponding techniques.

Q2: **[Time: 20 Minutes] [4+1 = 5 Marks]**

A. A game is played between MAX and MIN. Draw a tree considering MAX as the first player and branching factor 2. Given terminal values below, show backed-up values, the best decision available at the root, and branches that will get pruned.
   4, 5, 2, 3, 7, 6, 11, 2, -14, -7, -21, -24, 9, 10, 11, 12.
   Explain which processing of nodes either left-to-right or right-to-left would lead to an increase, decrease, or no change in the number of pruned branches.

B. How does the minimax algorithm improve decision-making within adversarial search scenarios?

CLO # 3: To demonstrate understanding and ability to implement the major concepts, approaches and research in evolutionary algorithms and CSP.

Q3: **[Time: 25 Minutes] [3+2 = 5 Marks]**

A. A group of three friends is planning their vacation trip for a week, and they have a list of five possible destinations to visit. Each friend can only join one destination per morning. The destinations and their corresponding time slots are:

    Hiking Trail: Opens from 1:00 pm-2:00 pm
    Beach Resort: Opens from 1:30 pm-2:30 pm
    Historical Museum: Opens from 2:00 pm-3:00 pm
    Amusement Park: Opens from 2:00 pm-3:00 pm
    Wildlife Safari: Opens from 2:30 pm-3:30 pm

   The friends who will be joining the trip are:

    Friend Alice, who is available to visit Destinations 3 and 4.
    Friend Bob, who is available to visit Destinations 2, 3, 4, and 5.
    Friend Carol, who is available to visit Destinations 1, 2, 3, 4, 5.

   Formulate this as a CSP problem by answering the following:

   1. Identify variables, domains, and constraints of the problem.
   2. Draw a constraint graph of the problem.
   3. Show the domains of the variables after running arc consistency on this initial graph and give one solution to this CSP.

B. You are solving the puzzle of cryptarithmetic that represents a different digit and none of the numbers use leading zeros. You have solved the puzzle of BLACK + GREEN = ORANGE and CROSS + ROADS = DANGER and got their numerical values.
   A remover nearby fell on your answer and although you got the solution, some values of your answers were erased, _2 _7 _ 1 and 1 _ _ 6 _ 4. Now solve again to get the correct answer.